

Automated and Explainable Outcome Prediction in NSCLC: Beyond Manual Feature Extraction using LLM Embeddings

Alberto Ferrarin¹, Teresa Arangoa², Cristina Licciardelo³, Vittoria Peppoloni¹, Laura Mazzeo¹, Giuseppe Leone¹, Giulia Montelatici¹, Cecilia Silvestri¹, Leonardo Provenzano¹, Giulia Corrao¹, Monica Ganzinelli¹, Andrea Spagnoletti¹, Teresa Beninato¹, Mario Occhipinti¹, Marta Brambilla¹, Claudia Proto¹, Giuseppe Lo Russo¹, Paolo Baili¹, Vanja Miskovic³, Federica Corso¹, Arsela Prelaj¹

1. Medical Oncology Department 1, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; 2. Medical Oncology Department, Hospital Universitario de La Princesa, Madrid, Spain; 3. Department of Electronic, Information and Bioengineering, Politecnico di Milano, Milan, Italy

INTRODUCTION

- **Clinical notes** contain rich, unstructured **prognostic information**.
- **Manual extraction** is labor-intensive, slow, and **not scalable** for large cohorts.
- **LLM embeddings** can automate this process by generating **context-aware features**.
- **Explainability** is essential to translate AI predictions into **trustworthy clinical insights**.

GOALS

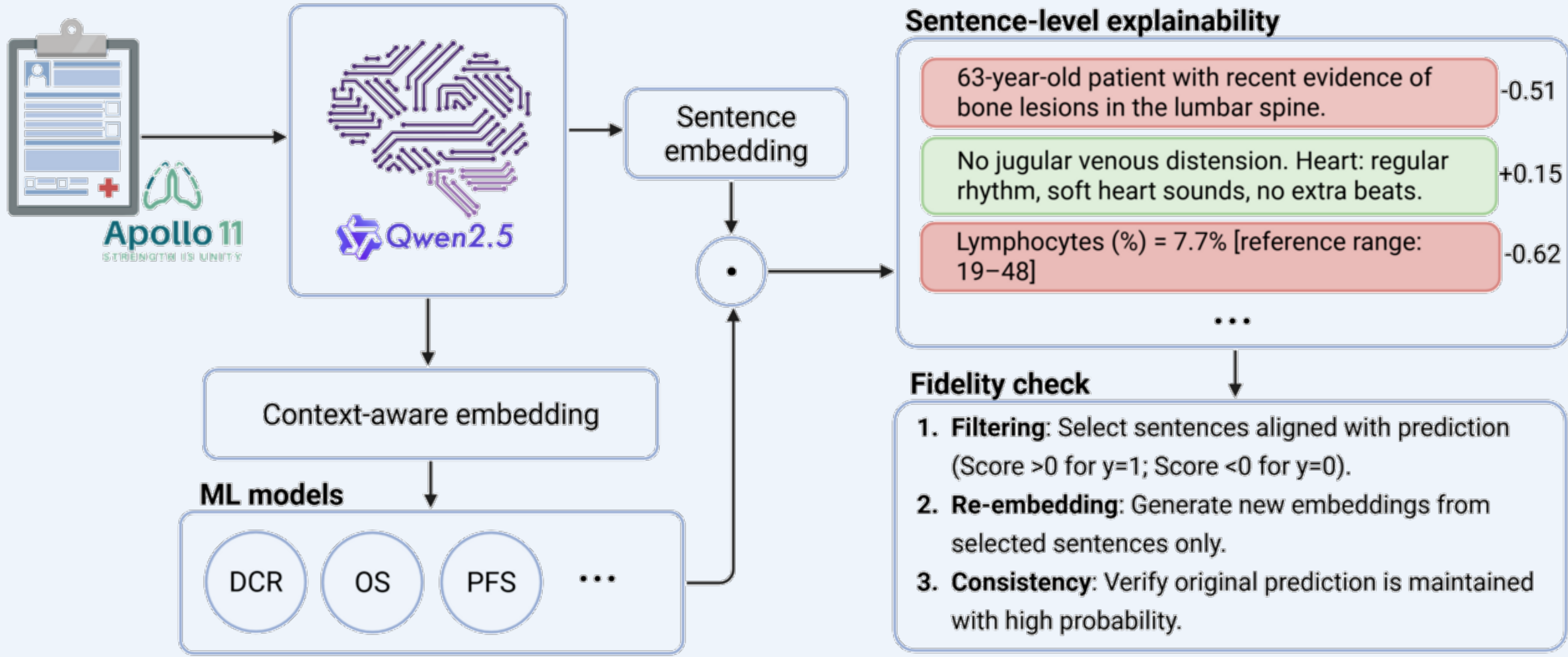
- Automate **feature engineering** from **unstructured notes**.
- Predict **OS, PFS, DCR**, and **toxicity** across **IO, CHT**, and **IO+CHT**.
- Provide **sentence-level explainability** for clinical transparency.
- **Validate** AI logic via a rigorous **Fidelity framework**.

DATASET

- **691 NSCLC patients**
 - Istituto Nazionale dei Tumori
- **Apollo 11** project
 - NCT05550961
- **Heterogeneous cohort** including patients treated with
 - **Immunotherapy**
 - **Chemotherapy**
 - **Immuno-Chemotherapy**
- Multimodal analysis of
 - unstructured **clinical reports**
 - **laboratory results**
 - **genomic profiling**

METHODS

- **Qwen2.5-7B** generates patient-level embeddings
- Implemented on a proprietary **on-premise server** to ensure data privacy and **GDPR compliance**
- **Embeddings** → **prediction models**
- Sentence-level **explainability** + **fidelity**



RESULTS

- Accurate prediction of **OS, PFS, DCR, toxicity**
- Explainability → key **clinical drivers**
- **Fidelity** validated explanations
- Fully **automated pipeline**

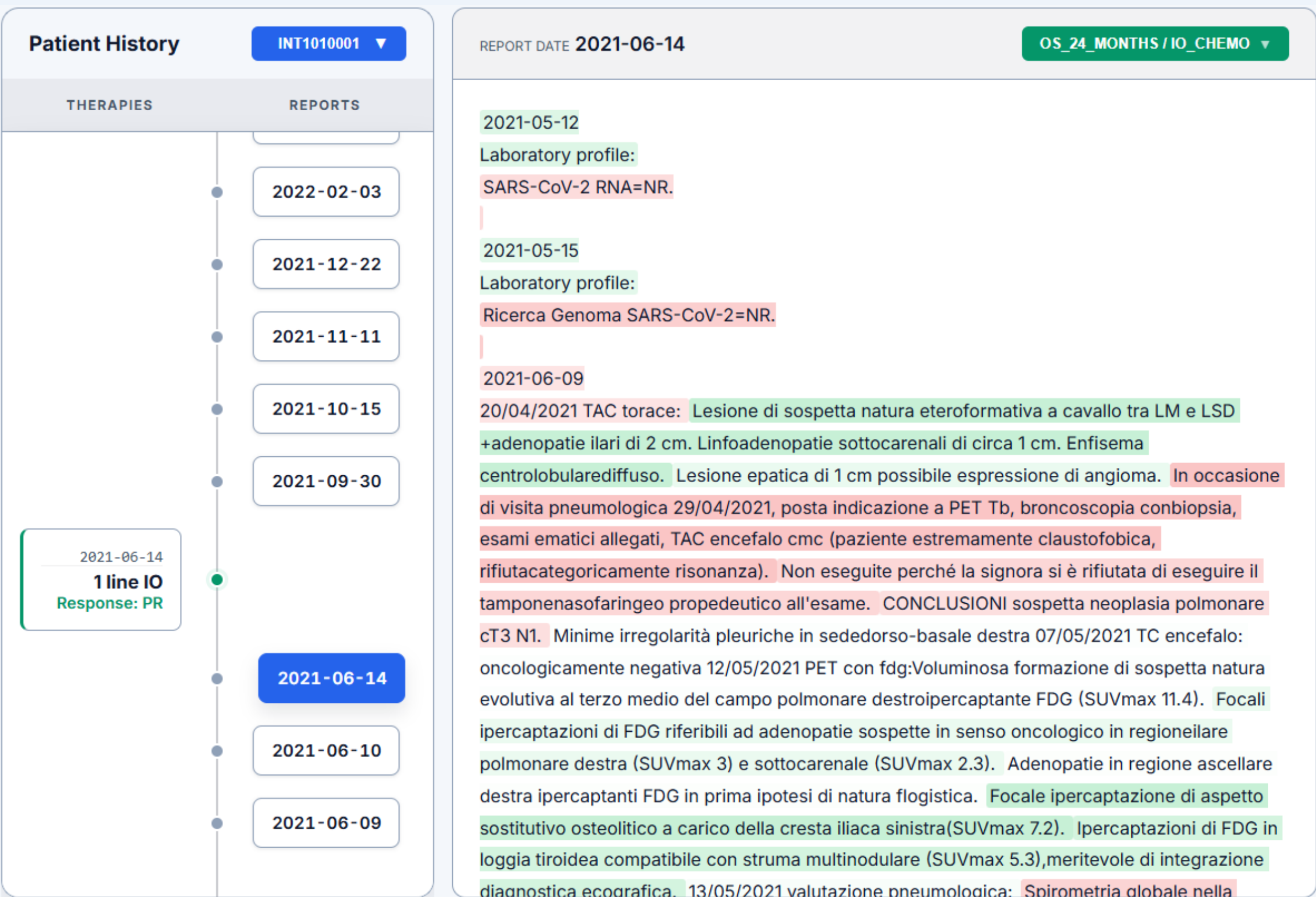
Outcome	IO	CHT	IO+CHT
OS 6	0.63 ± 0.13	0.79 ± 0.19	0.72 ± 0.20
OS 24	0.67 ± 0.16	0.81 ± 0.19	0.62 ± 0.22
DCR	0.55 ± 0.14	0.73 ± 0.19	0.56 ± 0.21
TOXICITY	0.57 ± 0.15	*	0.57 ± 0.24

Table 1: 5-fold CV AUC (Mean ± CI)

Outcome	IO	CHT	IO+CHT
OS	0.58 ± 0.04	0.68 ± 0.06	0.62 ± 0.06
PFS	0.61 ± 0.08	*	0.85 ± 0.10

Table 2: 5-fold CV C-index (Mean ± CI)

* Not assessed due to insufficient sample size for robust training.



Sentence-level explainability (OS 24, IO+CHT arm): green highlights favorable clinical factors, red indicates unfavorable ones.

CONCLUSION

- **LLM embeddings** replace manual feature engineering.
- **Transparent and reliable** clinical predictions.
- **Scalable** for real-world oncology.

Next Steps:

- **Longitudinal Analysis:** Capture disease evolution
- **Digital Twin:** Dynamic patient simulation



Project details & contacts

